

Employee Attrition Prediction using Machine Learning

Irfan Rahman, 24301330, and Zarifah Morshed Nuren, 24301499
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Abstract—Employee turnover poses a significant financial and operational challenge to modern organizations. This project develops a machine learning pipeline for predicting employee attrition using a comprehensive HR dataset containing 1,677 records. Exploratory data analysis was conducted to examine feature distributions, class imbalance, correlations, and potential predictors of attrition. Multiple supervised and unsupervised learning approaches were then implemented and evaluated using accuracy, precision, recall, F1-score, and AUC. Logistic Regression achieved the highest overall accuracy (88.1%) and AUC (0.77), while Gaussian Naive Bayes achieved the highest recall (0.55) for the minority “Left” class. The results demonstrate the importance of selecting evaluation metrics according to the practical objective of an imbalanced classification problem.

Index Terms—Machine Learning, Employee Attrition, Exploratory Data Analysis, Classification, Imbalanced Data

I. INTRODUCTION

EMPLOYEE turnover poses a major financial and operational challenge for organizations. This project aims to develop a machine learning pipeline for predicting employee attrition using HR-related features, including demographics, job roles, and job satisfaction. By analyzing patterns associated with employee turnover, the project explores how machine learning can support the identification of employees who may be at risk of leaving.

For this classification task, **Attrition** was selected as the target variable, where 0 represents employees who stayed and 1 represents employees who left. The project aims to bridge the gap between theoretical machine learning concepts and practical application by applying exploratory data analysis, preprocessing, and multiple supervised and unsupervised learning techniques to a real-world HR dataset. The performance of these approaches is subsequently compared using evaluation metrics appropriate for the imbalanced attrition prediction problem.

II. DATASET DESCRIPTION

The dataset used for this project contains comprehensive HR analytics data designed for employee attrition prediction. Based on the initial data exploration, the dataset comprises the following structural characteristics:

- **Data Points & Features:** The dataset contains 1,677 observations (rows) across 35 distinct features (columns), including the target variable `Attrition` and a unique identifier `id`.
- **Numerical Features:** There are 27 numerical features (data type `int64`), which include continuous and discrete variables such as `Age`, `MonthlyIncome`, and `DistanceFromHome`.

- **Categorical Features:** There are 8 categorical features (data type `object`), which include text-based variables such as `Gender`, `Department`, and `JobRole`.

III. DATA ANALYSIS

Prior to model training, Exploratory Data Analysis (EDA) was conducted to uncover patterns, evaluate feature distributions, and justify preprocessing decisions. Key insights include:

- **Data Quality & Cleanliness:** Inspection revealed zero missing values and no duplicate records. However, three features (`EmployeeCount`, `StandardHours`, `Over18`) contained only a single unique value. Lacking predictive variance, these constant features were identified for removal.
- **Class Imbalance:** The target feature exhibits significant imbalance (Fig. 3.1). Out of 1,677 records, 1,477 (88.07%) stayed while only 200 (11.93%) left. This 88-to-12 split rules out raw accuracy as a reliable metric, necessitating the use of Precision, Recall, F1-Score, and AUC for robust model evaluation.

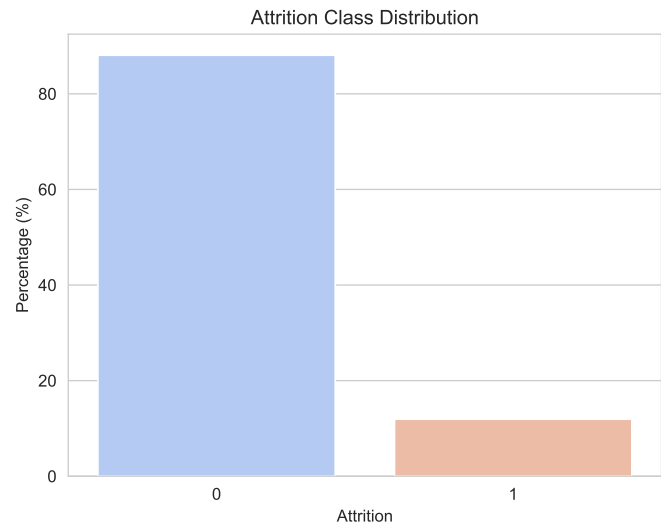


Fig. 3.1. Distribution of Employee Attrition (Stayed vs. Left)

- **Feature Skewness & Outliers:** Statistical evaluation of the numerical variables demonstrated notable right-skewness across several tenure and compensation metrics. As shown in Fig. 3.2, `YearsSinceLastPromotion` (2.08), `YearsAtCompany` (1.74), `MonthlyIncome`

(1.55), and TotalWorkingYears (1.15) all show a long right tail with visible outlier points beyond the upper whisker, which correspond to a small group of highly tenured, highly promoted, or highly compensated employees. Notably, Attrition itself showed the highest skew (2.35), though this stems from class imbalance rather than outliers, as discussed above and shown in Fig. 3.1.

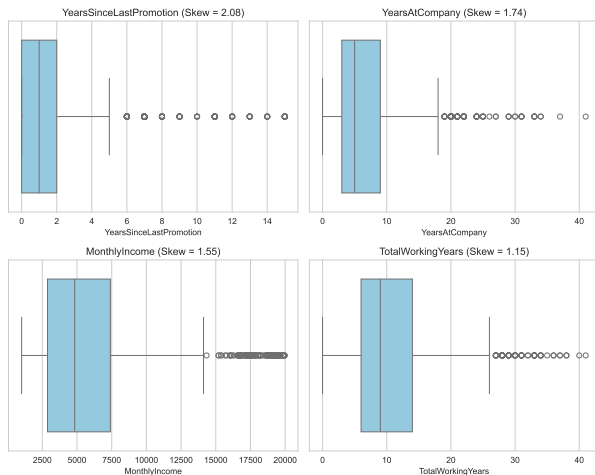


Fig. 3.2. Boxplots of Right-Skewed Numerical Features Showing Outliers

- **Key Attrition Drivers (Correlations & Trends):** Both continuous and categorical analyses isolated the most powerful predictors of employee turnover:

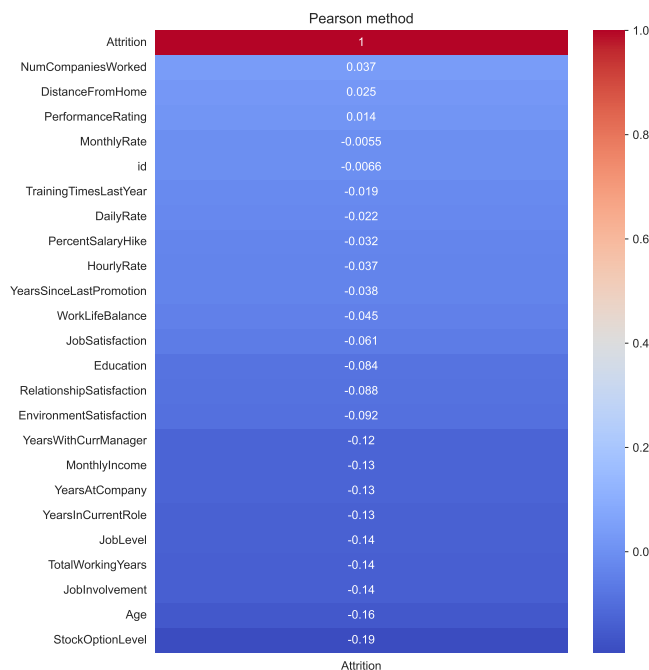


Fig. 3.3. Pearson Correlation of Numerical Features against Attrition

- **Negative Numerical Correlation:** As illustrated in Fig. 3.3, continuous features like StockOptionLevel (-0.19) and Age (-0.16) showed the strongest negative correlation with

attrition, implying that older, more financially invested employees are the least likely to leave.

- **Categorical Risk Factors:** Cross-tabulation (Fig. 3.4) revealed stark contrasts among discrete groups. Employees working OverTime depart at heavily inflated rates compared to those who do not. Furthermore, employees who are Single and those operating within the Sales Representative role display the highest relative turnover risks in their respective categories. This was quantitatively confirmed via Cramér's V (Fig. 3.5), which ranked JobRole (0.217), MaritalStatus (0.181), and OverTime (0.172) as the categorical features with the strongest association to Attrition, consistent with the visual patterns observed above.

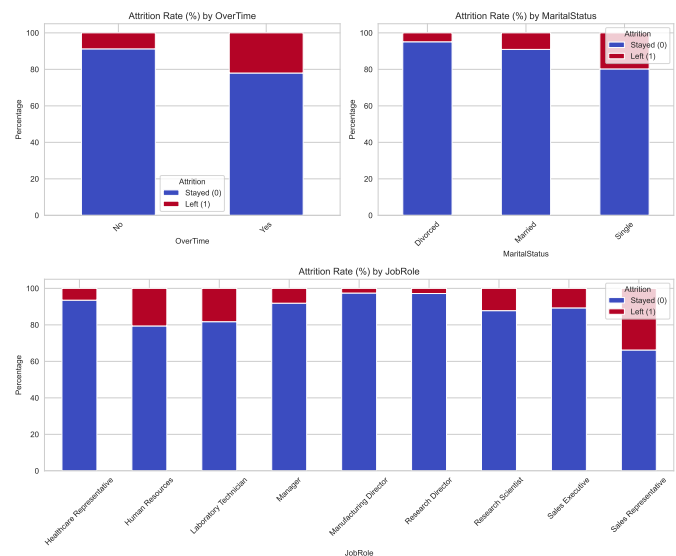


Fig. 3.4. Attrition Rates by Key Categorical Features

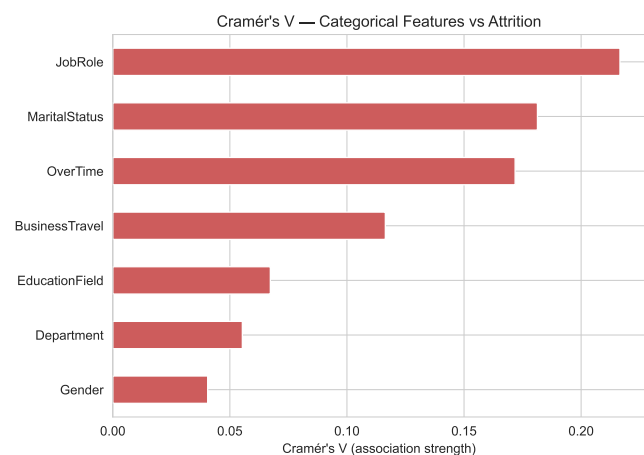


Fig. 3.5. Association Strength of Categorical Features with Attrition

- **Multicollinearity:** As shown in Fig. 3.6, several numerical features exhibit strong pairwise correlation. Following

common practice, $|r| > 0.9$ was treated as severe collinearity and $|r| > 0.7$ as strong correlation warranting attention. Most notably, `JobLevel` and `MonthlyIncome` are severely collinear (0.91), indicating near-total redundancy. A second cluster is visible among tenure-related features: `YearsAtCompany`, `YearsInCurrentRole`, and `YearsWithCurrManager` are all mutually correlated (0.75–0.79), reflecting that employees who stay long at the company also tend to stay long in their current role and under the same manager. This redundancy indicates that retaining all correlated features could negatively impact model performance and introduce unnecessary noise, directly motivating targeted feature selection during the preprocessing phase.

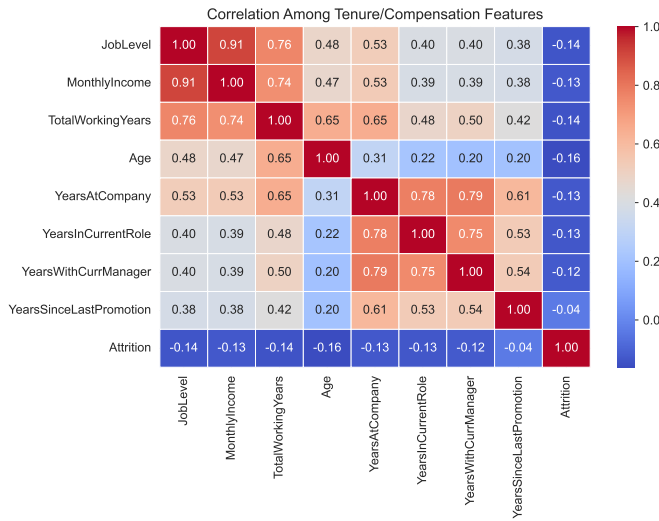


Fig. 3.6. Correlation Heatmap of Tenure and Compensation Features

Collectively, these findings (data cleanliness, class imbalance, skewed distributions, and feature multicollinearity) directly inform the preprocessing strategy detailed in the following section.

IV. DATASET PRE-PROCESSING

Building directly upon the findings of the exploratory data analysis presented in Section III, a systematic pre-processing pipeline was applied to prepare the dataset for model training. Each subsection below identifies an issue surfaced during the exploratory analysis and details the corresponding technique applied to resolve it.

A. Handling Null and Duplicate Values

Missing or duplicate records, if present, could introduce bias, reduce the amount of usable training data, or artificially inflate performance metrics through redundant observations. Section III established that the dataset contains zero missing values and zero duplicate records across all 1,677 observations; both conditions were independently re-verified immediately prior to preprocessing using `dataset.isnull().sum()` and `dataset.duplicated().sum()`. As both counts

remained zero, no imputation, row removal, or deduplication was required.

B. Encoding Categorical Values

Machine learning algorithms require numerical input, making the dataset's eight categorical features unsuitable for direct model training. Applying a single encoding strategy across all of them, however, risks imposing artificial ordinal relationships on features that possess no inherent ranking. To address this, each categorical feature was encoded according to its underlying semantic structure. The binary features `OverTime` and `Gender` were mapped directly to 0 and 1. `BusinessTravel` possesses a genuine ordering of travel frequency (`Non-Travel` < `Travel_Rarely` < `Travel_Frequently`) and was therefore encoded ordinally (0, 1, 2) to preserve this ranking. The remaining nominal features (`Department`, `EducationField`, `JobRole`, and `MaritalStatus`) exhibit no intrinsic order and were transformed using one-hot encoding, with the first category of each dropped to avoid the dummy variable trap. This produced 43 fully numerical columns, including the target variable; therefore, 42 predictor features remained after separating the target.

C. Feature Selection

The exploratory analysis identified two categories of features likely to negatively affect model performance: features carrying no predictive information, and features exhibiting severe multicollinearity. Accordingly, the unique identifier `id` and the three zero-variance features (`EmployeeCount`, `StandardHours`, `Over18`) were removed, as each contains only a single repeated value across all records and therefore no useful signal for predicting attrition. In addition, the severe collinearity identified between `JobLevel` and `MonthlyIncome` (Fig. 3.6, $r = 0.91$) was addressed by removing `JobLevel` while retaining `MonthlyIncome`, which provides a finer-grained, continuous representation of employee compensation. The secondary correlated cluster (`YearsAtCompany`, `YearsInCurrentRole`, and `YearsWithCurrManager`, $r = 0.75$ – 0.79) was intentionally retained, as each feature captures a temporally distinct aspect of employee tenure, and tree-based models such as Decision Trees are comparatively robust to moderate multicollinearity among predictors.

D. Feature Scaling

Section III (Fig. 3.2) revealed substantial right-skewness and visible outliers in several numerical features, particularly `MonthlyIncome` (skew = 1.55) and `YearsSinceLastPromotion` (skew = 2.08). Combined with the fact that numerical features span markedly different value ranges, these characteristics risk disproportionately influencing distance-based learning algorithms, as well as any scaling technique that relies on the mean or the full range of the data. Prior to scaling, the dataset was split into training (80%) and testing (20%) subsets using stratified sampling on the target variable, preserving the original 88.07%/11.93%

class distribution in both partitions; the scaler was subsequently fit exclusively on the training set and applied to both sets to prevent information leakage.

Three scaling techniques were evaluated. `MinMaxScaler` was rejected, as it rescales values relative to the observed minimum and maximum, making it highly sensitive to the extreme values already identified. `StandardScaler` was also rejected, since it standardizes using the mean and standard deviation, both of which are shifted by a small number of extreme observations, compressing the effective spread of the remaining majority of the data. `RobustScaler` was ultimately selected, as it centers and scales each feature using the median and interquartile range (IQR) rather than the mean and standard deviation, making the transformation resistant to the outliers identified in Fig. 3.2 and ensuring that a small number of highly tenured or highly compensated employees do not distort the scale for the remaining workforce.

TABLE I
CUMULATIVE EFFECT OF PRE-PROCESSING STEPS ON DATASET SHAPE

Step	Action	Shape
Initial	Raw dataset loaded	(1677, 35)
(C)	Dropped <code>id</code> , constants	(1677, 31)
(B)	Categorical encoding	(1677, 43)
(C)	Dropped <code>JobLevel</code>	(1677, 42)
Split	80/20 train-test (stratified)	Train (1341, 42) Test (336, 42)

V. MODEL TRAINING

Building on the pre-processed dataset established in Section IV, four models were trained to examine the employee attrition problem from complementary perspectives. A feedforward Neural Network was implemented as the primary model, alongside two supervised classifiers, Logistic Regression and Gaussian Naive Bayes, for comparative analysis in Section VI. In addition, K-Means clustering was applied as an unsupervised approach to examine whether employees could be grouped according to similarity in their feature profiles without using the `Attrition` labels. Each subsection describes the configuration and training procedure of the respective model.

A. Neural Network

The primary model was a feedforward neural network implemented using TensorFlow/Keras. The architecture consists of a single hidden layer with 128 neurons using ReLU activation, followed by an output layer with a single neuron using sigmoid activation, suitable for binary classification. The model was compiled with the Adam optimizer (learning rate = 0.001) and binary cross-entropy loss, which is appropriate given the binary nature of the target variable and its probabilistic interpretation of the sigmoid output. The model was trained for 10 epochs with a batch size of 32 on the scaled training set (1,341 samples, 42 features).

Fig. 5.1 shows the training loss curve over the 10 epochs. The loss decreases sharply within the first few epochs, from approximately 0.52 to below 0.30 by epoch 4, before gradually

flattening out toward 0.24 by the final epoch. This smooth decline indicates stable training without signs of divergence or erratic fluctuation. Evaluation on the test set yielded an accuracy of 88.39% and a loss of 0.3246. A more complete evaluation using precision, recall, F1-score, and AUC is presented in Section VI.

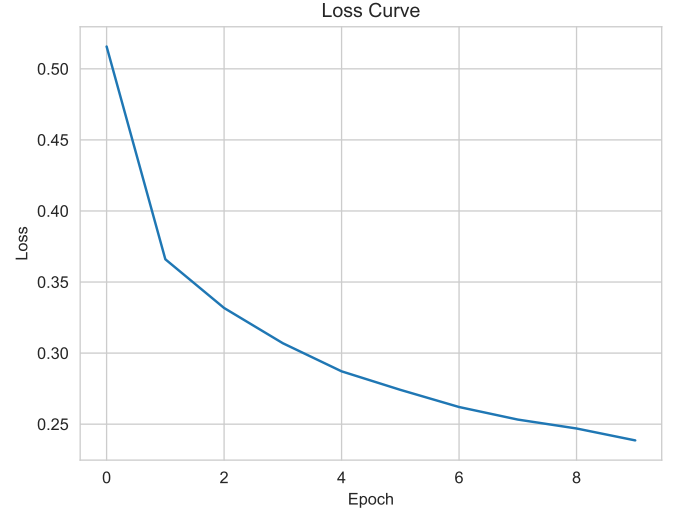


Fig. 5.1. Training Loss Curve for the Neural Network

B. Logistic Regression

As a baseline linear classifier, a Logistic Regression model was trained using scikit-learn's default configuration on the same scaled training data. Logistic Regression was selected as a comparative baseline due to its interpretability and its effectiveness as a benchmark for binary classification tasks. The model achieved a training accuracy of 89.63% and a test accuracy of 88.1%, results that are comparable to those of the neural network. This similarity in raw accuracy is not surprising given the class imbalance discussed earlier, and is further explored using more robust metrics in Section VI.

C. Gaussian Naive Bayes

A Gaussian Naive Bayes classifier was also trained to provide a probabilistic comparison point. This model assumes conditional independence between features given the class label and models each numerical feature as following a Gaussian distribution. The Gaussian Naive Bayes model achieved a training accuracy of 71.81% and a test accuracy of 68.15%, notably lower than both the neural network and logistic regression models. This lower performance may be partly explained by the model's conditional-independence assumption, which is not fully consistent with several correlated features identified during EDA (Section III), such as the tenure-related cluster (`YearsAtCompany`, `YearsInCurrentRole`, `YearsWithCurrManager`) and the one-hot encoded categorical dummy variables, which are not truly Gaussian-distributed.

D. K-Means Clustering (Unsupervised)

To examine the dataset from an unsupervised perspective, a K-Means clustering algorithm was applied to the scaled training features with $k = 2$ clusters, chosen to provide a comparable two-group structure to the binary attrition problem, and 10 different centroid initializations ($n_init = 10$) to improve stability of the final clustering solution. Unlike the supervised models, K-Means was not provided with the `Attrition` labels during training; instead, it grouped employees purely based on similarity across the 42 scaled features.

VI. MODEL EVALUATION

Following the training of the four models described in Section V, each model's predictive performance was assessed on the held-out test set of 336 employees (296 who stayed and 40 who left) using four complementary evaluation methods: confusion matrices, classification reports, ROC curves with AUC scores, and a summary comparison table. Because K-Means clustering is unsupervised, its two output clusters do not inherently correspond to the "Stayed" or "Left" classes. To make its output comparable to the supervised models, the attrition rate of each cluster on the **training set** was calculated, and the cluster with the higher attrition rate was labeled "Left." This mapping was then applied to the test set predictions to evaluate how well the unsupervised grouping aligned with the true labels; the K-Means algorithm itself did not use the `Attrition` column during clustering.

A. Confusion Matrices

The confusion matrices reveal a pattern strongly influenced by the class imbalance identified in Section III. The three supervised models generally identified employees who stayed much more accurately than those who left, although the degree of this imbalance varied substantially between models. The Neural Network and Logistic Regression correctly identified 290 and 288 of the 296 "Stayed" cases, respectively, but identified only 5 and 8 of the 40 "Left" cases, missing the majority of actual leavers.

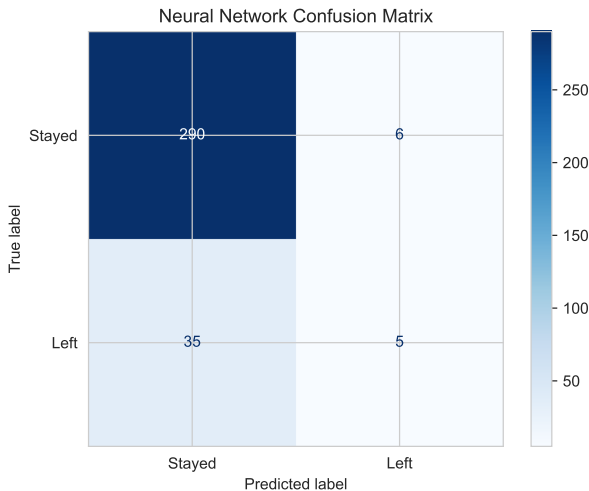


Fig. 6.1. Confusion Matrix for the Neural Network

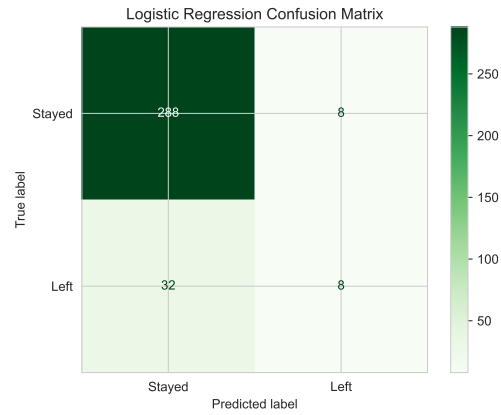


Fig. 6.2. Confusion Matrix for the Logistic Regression Model

Gaussian Naive Bayes shifted this trade-off considerably, correctly identifying 22 of the 40 "Left" cases at the cost of misclassifying 89 "Stayed" employees as leavers.

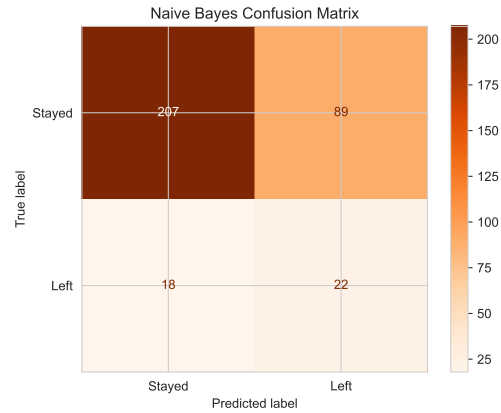


Fig. 6.3. Confusion Matrix for the Gaussian Naive Bayes Model

K-Means exhibited the most extreme trade-off. It correctly identified 33 of the 40 employees who left, but predicted "Left" for 223 of the 336 employees overall, resulting in 223 false positives among the "Stayed" group.

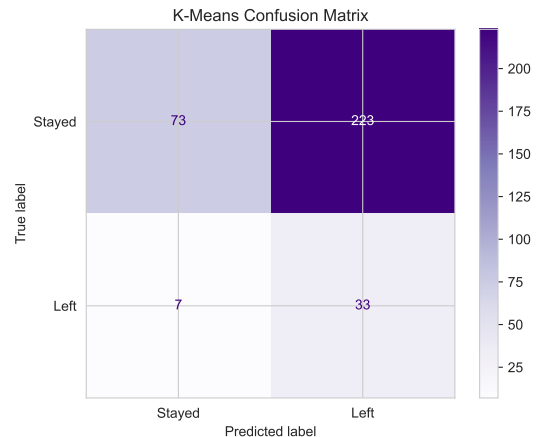


Fig. 6.4. Confusion Matrix for K-Means Clustering

B. Classification Reports

The per-class precision, recall, and F1-scores further quantify these differences. For the minority “Left” class, the Neural Network achieved a precision of 0.45 and recall of 0.12 (F1 = 0.20), while Logistic Regression achieved a precision of 0.50 and recall of 0.20 (F1 = 0.29). These results indicate that both models are strongly biased toward the majority class. Gaussian Naive Bayes traded precision for recall in this dataset, achieving a precision of 0.20 and recall of 0.55 (F1 = 0.29) for the “Left” class, making it more suitable when identifying a larger proportion of at-risk employees is prioritized over minimizing false alarms. K-Means achieved the highest recall of all four approaches for the “Left” class (0.825), but its precision was only 0.13, meaning that most employees predicted as “Left” were actually from the “Stayed” class. This reflects the large number of false positives produced by the unsupervised clustering approach.

C. ROC Curves and AUC

The ROC curves provide a threshold-independent view of each model’s discriminative ability. The Neural Network and Logistic Regression produced nearly identical curves, each achieving an AUC of 0.77, indicating similar overall ability to distinguish employees likely to leave from those likely to stay. Gaussian Naive Bayes achieved a slightly lower AUC of 0.70. K-Means, evaluated using the signed distance to its cluster centroids as a continuous score, achieved an AUC of 0.57, only marginally above the 0.50 random-classification baseline. This indicates that the feature-similarity structure identified by K-Means provides limited discrimination of attrition outcomes, reinforcing the advantage of supervised learning when labeled attrition information is available.

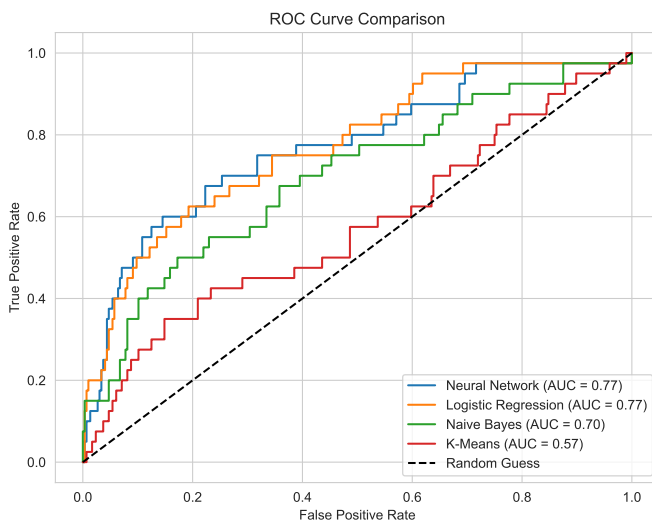


Fig. 6.5. ROC Curve Comparison Across All Four Models

D. Summary Comparison

Table II consolidates accuracy, precision, recall, F1-score, and AUC across all four models. Logistic Regression achieved

the highest overall accuracy (88.1%) and tied for the highest AUC (0.77), while the Neural Network achieved 87.8% accuracy. However, the high accuracy of both models is largely driven by their strong performance on the majority “Stayed” class, while their low recall for “Left” (0.125–0.200) limits their effectiveness when identifying at-risk employees is the primary objective.

Gaussian Naive Bayes achieved substantially higher recall for the “Left” class (0.550) than either Logistic Regression or the Neural Network, although this came at the cost of lower precision and overall accuracy (68.2%). Its minority-class F1-score of 0.291 was the highest among the supervised models, making it a useful alternative when detecting a greater proportion of potential leavers is prioritized.

K-Means showed the weakest overall discriminative performance, with an accuracy of 31.5% and an AUC of 0.570, despite achieving the highest “Left” recall (0.825). Its high recall was accompanied by an extremely low precision (0.129) and a large number of false positives. Overall, the results indicate that attrition does not form a cleanly separable structure based solely on unsupervised feature similarity, while supervised models benefit from direct access to the labeled attrition outcome.

TABLE II
PERFORMANCE COMPARISON ACROSS ALL MODELS

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.881	0.500	0.200	0.286	0.770
Neural Network	0.878	0.455	0.125	0.196	0.770
Gaussian Naive Bayes	0.682	0.198	0.550	0.291	0.700
K-Means Clustering	0.315	0.129	0.825	0.223	0.573

VII. CONCLUSION

Based on the evaluation presented in Section VI, Logistic Regression emerged as the strongest-performing model overall, achieving the highest accuracy (88.1%) and the highest AUC (0.77), while the Neural Network performed very similarly with 87.8% accuracy and an AUC of 0.77. The comparable performance of these two models suggests that the predictive patterns present in the dataset can be captured effectively by a relatively simple decision boundary, without requiring the additional complexity of the neural network.

However, accuracy alone does not provide a complete picture because of the substantial class imbalance in the dataset (88.07% “Stayed” versus 11.93% “Left”). Both Logistic Regression and the Neural Network achieved high accuracy while detecting relatively few employees who actually left, with recall values of 0.200 and 0.125, respectively. This demonstrates that a model can achieve apparently strong overall accuracy while still performing poorly on the minority class that is of greater practical interest in an employee attrition prediction problem.

When the objective is instead to identify as many potentially at-risk employees as possible, Gaussian Naive Bayes becomes a more practically useful alternative. Although it achieved lower accuracy (68.2%), it obtained a recall of 0.550 for the “Left” class, correctly identifying more than half of the employees who actually left. This came at the cost of lower precision

and a larger number of false positives, illustrating the trade-off between identifying more at-risk employees and avoiding unnecessary interventions.

K-Means performed considerably worse for the attrition prediction objective, achieving an accuracy of 31.5% and an AUC of 0.57. Its clustering results showed that the discovered groups did not correspond cleanly to the actual attrition outcomes, demonstrating the limitation of relying solely on unsupervised feature similarity for this particular task.

Overall, there is no single model that can be considered unambiguously “best” without first defining the objective. Logistic Regression is the strongest choice based on overall accuracy and AUC, whereas Gaussian Naive Bayes is more suitable when the priority is to identify a larger proportion of employees at risk of leaving. The project therefore highlights an important consideration in imbalanced classification: **model performance should be evaluated using metrics that reflect the actual objective of the prediction task, rather than relying on accuracy alone.**